BITTRUTH : To Verify Authenticity of Media

Problem Statement

The manual verificationofmedia isunreliable and often fails to detect sophisticated deepfakes. Existing platforms lack the AI-based tools necessary to identify subtle facial inconsistencies or signs of digital manipulation. Consequently, there is an urgent need for an automated machine learning system that provides accurate confidence scores to bridge the misinformation gap and reduce the spread of fake news.

Objectives

The project aims to developan automated machine learning system that analyzes media for facial inconsistencies to accurately classify it as "Real" or "Fake". By providing clear confidence scores through a user-friendly interface, the system ensures information integrity and helps non- technical users identify deepfakes before they are shared. This approach ultimately empowers individuals to verify the authenticity of digital content .

Key References

DeepfakeDetection Challenge(DFDC)FaceForensics++

Vision Transformer (ViT)

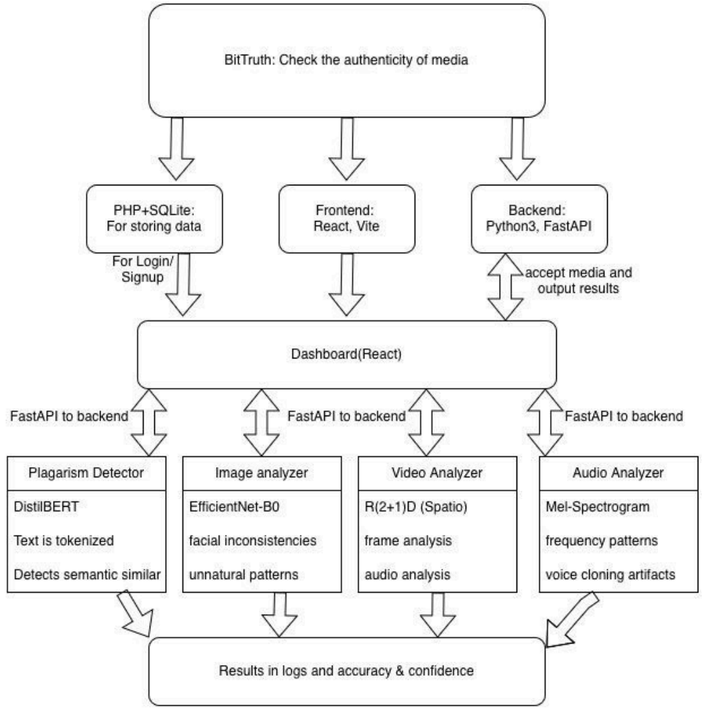
Research TIJER Publication

IJCRT Research

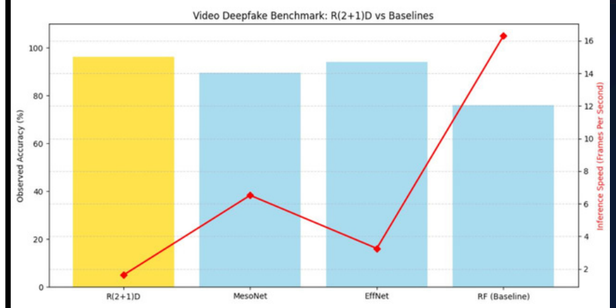
Conclusion

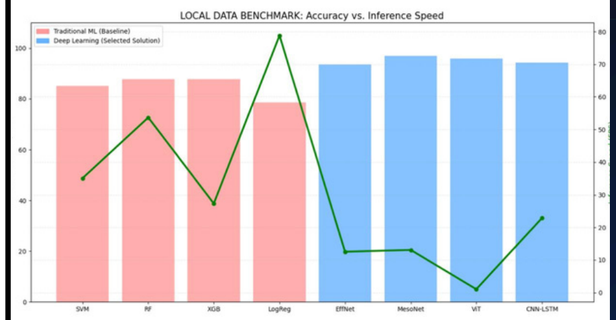
BitTruth is an advanced AI system that successfully detects deepfakes in images and videos by integrating XceptionNet and Vision Transformers for high reliability. It provides a practical tool for combating fraud through clear authenticity verdicts and is designed to scale for real-time verification against evolving digital manipulation.

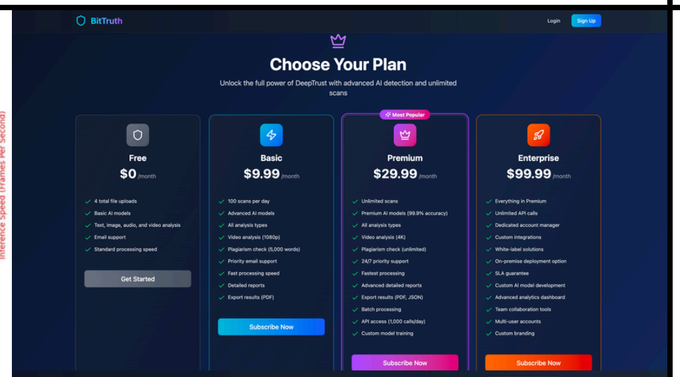
Architecture/Block Diagram

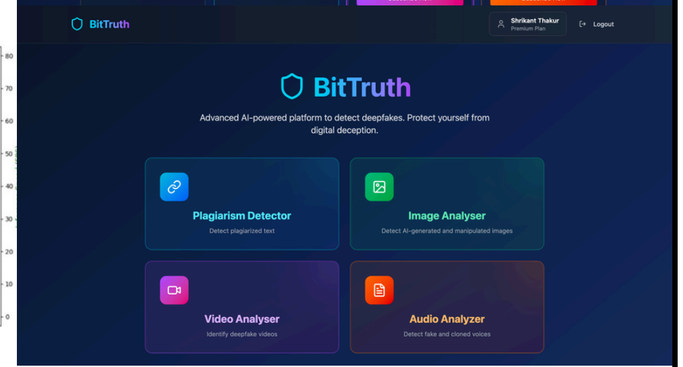
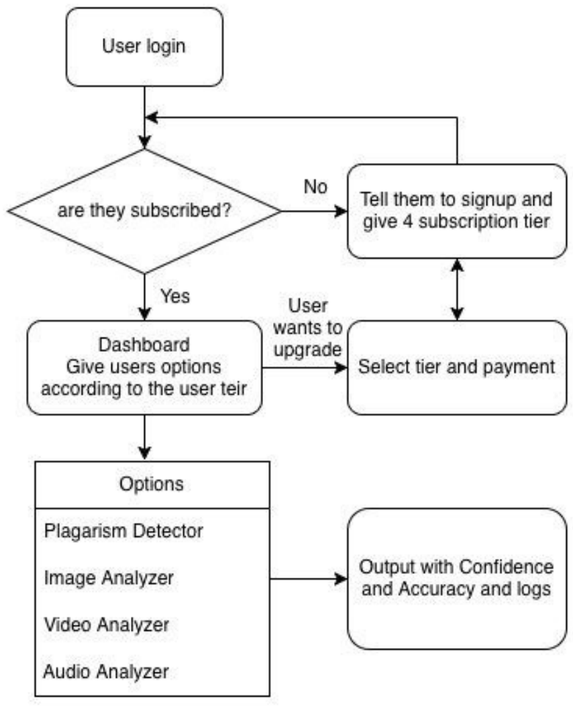


Results









Workflow